

Full local stability of the Gaussian propeller

Run `fa_banach_001`, arXiv:1112.2993, partial result

Source problem

For a centered finite list $v_1, \dots, v_k$ in a real Hilbert space, let G be standard Gaussian on its span and set

$$\mathcal{Q}(v_1, \dots, v_k) = \frac{(\mathbb{E} \max_{1 \leq i \leq k} \langle G, v_i \rangle)^2}{\sum_{i=1}^k \|v_i\|^2}.$$

The Gaussian formulation of the Propeller Conjecture asks whether

$$\mathcal{Q}(v_1, \dots, v_k) \leq c_0, \quad c_0 = \frac{9}{8\pi}, \quad (1)$$

for every k and every ambient dimension; see Khot–Naor [1] and Heilman–Jagannath–Naor [2]. The latter paper proves the three-dimensional case. On page 7 it explicitly identifies a proof extending to all dimensions as the next natural step.

errors. We remark that our arguments extend mutatis mutandis to higher dimensions. For the sake of simplicity we present the entire argument in $\mathbb{R}^3$, since while it is conceivable that the scheme presented here can yield a computer-assisted proof of the Propeller Conjecture in higher dimensions, at some fixed dimension the computer-assisted component of the proof will become unfeasible. Now that we know that a nonintuitive case of the Propeller Conjecture is indeed correct, the next natural step is to search for a proof that extends to all dimensions, rather than attempting to prove a few more cases in low dimensions.

2. AN OVERVIEW OF THE PROOF OF THEOREM [1.1](#)

The global problem remains open. We prove that the conjectured optimizer is locally rigid under arbitrary higher-dimensional perturbations, substantially strengthening the previously isolated purely transverse case.

Proof intuition

Choose the three vertices that are close to the propeller blades and translate the entire list by their mean. Translation does not change the expected Gaussian maximum. The translated list consists of a centered active triangle and a small arbitrary residual cloud. Re-centering the whole list gives an exact residual denominator cost of order η^2 .

The support function of a nondegenerate centered triangle dominates a fixed multiple of the planar radius. A residual vertex of total size η can win only when the planar Gaussian component is unusually small compared with its orthogonal component. The small-ball probability of a two-dimensional Gaussian is quadratic, so the expected improvement is only $O(\eta^3)$. Thus the quadratic denominator cost beats the cubic numerator gain.

The active triangle needs no external sharp inequality. Cauchy’s planar perimeter formula expresses its expected Gaussian support in terms of its perimeter, and one application of Cauchy–Schwarz gives the sharp constant $9/(8\pi)$, with equality exactly for an equilateral triangle.

Quantitative theorem

The following statement is useful beyond a neighborhood of the equilateral triangle. It gives an explicit certificate for adjoining an arbitrary small cloud to any nondegenerate centered triangle.

Theorem 1 (Arbitrary-cloud certificate). *Let u_1, u_2, u_3 lie in a finite-dimensional real Hilbert space, satisfy $u_1 + u_2 + u_3 = 0$, and span a two-dimensional space U . Put*

$$S = \sum_{i=1}^3 \|u_i\|^2, \quad M = \max_{1 \leq i \leq 3} \langle X, u_i \rangle, \quad W = \mathbb{E}M,$$

where X is standard Gaussian on U , and define the inradius parameter

$$\alpha = \min_{x \in U, \|x\|=1} \max_{1 \leq i \leq 3} \langle x, u_i \rangle > 0.$$

Fix $m \geq 1$, set $k = m + 3$, and let $y_1, \dots, y_m$ be arbitrary vectors in the ambient Hilbert space. Write

$$\eta^2 = \sum_{j=1}^m \|y_j\|^2, \quad A = \eta^2 - \frac{1}{k} \left\| \sum_{j=1}^m y_j \right\|^2, \quad \bar{y} = \frac{1}{k} \sum_{j=1}^m y_j.$$

Form the centered list

$$v_i = u_i - \bar{y} \quad (1 \leq i \leq 3), \quad v_{3+j} = y_j - \bar{y} \quad (1 \leq j \leq m).$$

If $0 < \eta < \alpha$, define

$$b = \frac{\sqrt{2/\pi}}{2(\alpha - \eta)^2}.$$

Whenever

$$2Wb\eta^3 + b^2\eta^6 < c_0A, \tag{2}$$

one has

$$\mathcal{Q}(v_1, \dots, v_k) < c_0.$$

Moreover $A \geq (3/k)\eta^2$, so (2) holds for all sufficiently small nonzero η , uniformly over the directions and number of dimensions used by the residual cloud.

Corollary 2 (Full local stability). *Fix $k \geq 4$ and an equilateral centered triple u_1^0, u_2^0, u_3^0 . In the space of centered k -tuples there is a neighborhood of*

$$(u_1^0, u_2^0, u_3^0, 0, \dots, 0)$$

such that every tuple in this neighborhood satisfies (1). Equality holds precisely when, after the same labeling, the first three vectors form an equilateral centered triple and every remaining vector is zero. Thus, modulo rotations, scaling, and relabeling, the propeller is a strict local maximizer against all perturbations in arbitrary ambient dimension.

Proof

We first record the sharp estimate for the active triangle.

Lemma 3 (Centered-triangle inequality). *With the notation of Theorem 1,*

$$W^2 \leq c_0 S,$$

with equality if and only if the triangle with vertices u_1, u_2, u_3 is equilateral.

Proof. Let P be the perimeter of $K = \text{conv}\{u_1, u_2, u_3\}$. Write $X = R\Theta$, where Θ is uniform on the unit circle in U and R is independent of Θ . The planar Cauchy perimeter formula says that the angular average of the support function h_K is $P/(2\pi)$. Since $\mathbb{E}R = \sqrt{\pi/2}$,

$$W = \mathbb{E}h_K(X) = \frac{P}{2\sqrt{2\pi}}.$$

If a, b, c are the three side lengths, the centroid identity gives

$$a^2 + b^2 + c^2 = 3S.$$

Consequently

$$P^2 = (a + b + c)^2 \leq 3(a^2 + b^2 + c^2) = 9S,$$

and therefore $W^2 \leq 9S/(8\pi) = c_0 S$. Equality in Cauchy–Schwarz occurs exactly when $a = b = c$. $\square$

Proof of Theorem 1. The list $v_1, \dots, v_k$ is centered. Adding the common centered Gaussian $-\langle G, \bar{y} \rangle$ does not change the expectation of the maximum, so

$$\mathbb{E} \max_{1 \leq i \leq k} \langle G, v_i \rangle = \mathbb{E} \max \left\{ M, \max_{1 \leq j \leq m} \langle G, y_j \rangle \right\}. \quad (3)$$

The denominator has the exact identity

$$\sum_{i=1}^k \|v_i\|^2 = S + A. \quad (4)$$

Indeed, this is the standard variance identity after subtracting the mean $\bar{y}$. Since $\|\sum_j y_j\|^2 \leq m\eta^2$ and $k = m + 3$,

$$A \geq \left(1 - \frac{m}{k}\right) \eta^2 = \frac{3}{k} \eta^2. \quad (5)$$

Decompose $G = X + Z$ according to $U \oplus U^\perp$, and write $y_j = p_j + q_j$ in the same decomposition. Set

$$Q(Z) = \max_{1 \leq j \leq m} \langle Z, q_j \rangle.$$

Because $\|p_j\| \leq \eta$ and $M \geq \alpha \|X\|$, the improvement over the active triangle is bounded by

$$\begin{aligned} \delta &:= \mathbb{E} \left(\max \left\{ M, \max_j \langle G, y_j \rangle \right\} - M \right) \\ &\leq \mathbb{E} (Q(Z) - (\alpha - \eta) \|X\|)_+. \end{aligned} \quad (6)$$

For $q > 0$ and $a = \alpha - \eta > 0$, the radius of a standard two-dimensional Gaussian satisfies

$$\Pr\{\|X\| \leq t\} = 1 - e^{-t^2/2} \leq \frac{t^2}{2}.$$

Conditioning on Z therefore gives

$$\mathbb{E}_X(q - a\|X\|)_+ \leq q \Pr\{\|X\| \leq q/a\} \leq \frac{q^3}{2a^2}.$$

Also

$$\mathbb{E}Q(Z)_+^3 \leq \sum_{j=1}^m \mathbb{E}\langle Z, q_j \rangle_+^3 = \sqrt{\frac{2}{\pi}} \sum_{j=1}^m \|q_j\|^3 \leq \sqrt{\frac{2}{\pi}} \eta^3.$$

Combining this with (6) proves the uniform cubic bound

$$\delta \leq b\eta^3. \quad (7)$$

By Lemma 3, (3), (4), and (7),

$$\mathcal{Q}(v_1, \dots, v_k) \leq \frac{(W + b\eta^3)^2}{S + A}.$$

The numerator on the right is $W^2 + 2Wb\eta^3 + b^2\eta^6$. Lemma 3 bounds the first term by c_0S , while (2) bounds the last two strictly by c_0A . This proves the claim. The final assertion follows from (5), since the left side of (2) is $O(\eta^3)$ and its right side is bounded below by a positive multiple of η^2 . $\square$

Proof of Corollary 2. For a centered k -tuple $(z_1, \dots, z_k)$ near the displayed propeller, let $a = (z_1 + z_2 + z_3)/3$, put $u_i = z_i - a$ for $i \leq 3$, and put $y_j = z_{3+j} - a$. Subtracting a from the entire list does not change the expected maximum. The centered version of the resulting list is exactly the original tuple, so Theorem 1 applies.

In a sufficiently small neighborhood of the fixed equilateral triple, the parameters α are bounded below by some $\alpha_* > 0$ and the values W are bounded above by some $W_* < \infty$. If $\eta \leq \alpha_*/2$, then

$$b \leq B_* := \frac{2\sqrt{2/\pi}}{\alpha_*^2}.$$

Hence (2) follows uniformly once

$$2W_*B_*\eta + B_*^2\eta^4 < \frac{3c_0}{k},$$

which holds for all sufficiently small positive η . If $\eta > 0$, the conclusion is strict. If $\eta = 0$, all residual vectors vanish after the translation, and centeredness forces $a = 0$; Lemma 3 gives equality exactly for an equilateral active triple. This proves the stated local rigidity and equality characterization. $\square$

Verification notes

The proof is analytic and has no computational dependency. The two points most worth checking are the exact centered-denominator identity (4) and the passage from (6) to the cubic estimate (7). Both constants are deliberately kept explicit. The argument is dimension-free after the planar small-ball step: the estimate $\sum_j \|q_j\|^3 \leq (\sum_j \|q_j\|^2)^{3/2}$ removes any dependence on the number of transverse directions.

As a sanity check, for an equilateral triangle of circumradius r one has $\alpha = r/2$, $S = 3r^2$, and $W = 3\sqrt{3}r/(2\sqrt{2\pi})$, which gives $W^2/S = 9/(8\pi)$.

Scope and novelty check

This is a partial result, not a proof of the global higher-dimensional conjecture. It controls every sufficiently small perturbation of the three-blade optimizer for each fixed k , but it does not control configurations far from that optimizer.

A bounded primary-source search on 2026-08-09 used the exact phrases “Propeller Conjecture”, “higher dimensions”, “Gaussian moments”, and “local stability”, together with arXiv:0807.4626 and arXiv:1112.2993. It also inspected the primary arXiv records 1901.03934 and 2209.11216. The search found the known three-dimensional theorem and later conditional work, but no unconditional all-dimensional solution and no theorem matching the arbitrary-perturbation local statement above. Novelty confidence is moderate.

The earlier run packet `0807.4626_propeller_transverse_local_stability` proves only the special case where a fixed residual cloud is scaled in a space orthogonal to an undeformed equilateral triangle. The present theorem removes both restrictions and supplies a quantitative certificate. Human review is recommended, with priority on (4) and (7).

References

- [1] S. Khot and A. Naor, *Approximate kernel clustering*, arXiv:0807.4626, 2008.
- [2] S. Heilman, A. Jagannath, and A. Naor, *Solution of the propeller conjecture in $\mathbb{R}^3$* , arXiv:1112.2993, 2011.